COURSE CERTIFICATE

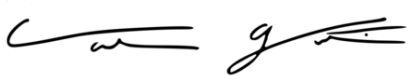
NOVEMBER 22, 2015

Alexander Sommer

has successfully completed

Machine Learning Foundations: A Case Study Approach

a MOOC from the University of Washington and offered through Coursera



Carlos Guestrin
Amazon Professor of Machine Learning
Computer Science and Engineering



Emily Fox
Amazon Professor of Machine Learning
Statistics



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Coursera has confirmed the identity of this individual and their participation in the course.



4 Courses

- Blockchain Basics
- Smart Contracts
- Decentralized Applications (Dapps)
- Blockchain Platforms

12/14/2018

Alexander Sommer

has successfully completed the online, non-credit Specialization

Blockchain

Through this specialization, learners developed an understanding of foundational concepts that enable a blockchain protocol. The courses covered applying the concepts of encryption, hashing, consensus, transactions, blocks and private-public keys in building a blockchain. Learners designed, developed and tested smart contracts and decentralized applications on a private Ethereum blockchain. The discussions included the architecture of a decentralized application stack, best practices, emerging standards, and many open issues such as scalability and privacy. Learning concluded with a holistic view of the landscape, including decentralized application use cases and other blockchain platforms.

Bina Ramamurthy Timothy Leyh

Bina Ramamurthy,
Teaching Professor of
the University at Buffalo
Computer Science and
Engineering Department

Timothy Leyh, Executive
Director of the University
at Buffalo Center for
Industrial Effectiveness

Verify this certificate at:
coursera.org/verify/specialization/YGNZ3A9ZX2QR

Duke
UNIVERSITY

COURSE
CERTIFICATE

08/08/2016

Alexander Sommer

has successfully completed

Visual Perception and the Brain

an online non-credit course authorized by Duke University and offered through Coursera



Professor Dale Purves, MD
Geller Professor of Neurobiology
Duke Institute for Brain Sciences
Duke University



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Coursera has confirmed the identity of this individual and their participation in the course.

COLUMBIA UNIVERSITY
BARNARD

VERIFIED
CERTIFICATE

AUGUST 03, 2015

Alexander Sommer

has successfully completed

Economics of Money and Banking, Part One

a 7 week online non-credit course authorized by Columbia University and offered through Coursera



Professor Perry Mehrling
Barnard College, Columbia University
Institute for New Economic Thinking



Verify at coursera.org/verify/F78XPWXYZK

Coursera has confirmed the identity of this individual and their participation in the course.

coursera

coursera.org

JUNE 21, 2015

Statement of Accomplishment

ALEXANDER SOMMER

HAS SUCCESSFULLY COMPLETED IE BUSINESS SCHOOL'S ONLINE OFFERING OF CRITICAL PERSPECTIVES ON MANAGEMENT.



Critical Perspectives on Management

This MBA-level course examines aspects of modern management strategy through a critical lens to examine their underlying foundations as well as their long-term outcomes, and provides an introduction to the conceptual and theoretical language used in the study of management.

PROFESSOR ROLF STROM-OLSEN
IE BUSINESS SCHOOL, MADRID

THE ONLINE OFFERING OF THIS CLASS DOES NOT REFLECT THE ENTIRE CURRICULUM OFFERED TO STUDENTS ENROLLED AT IE BUSINESS SCHOOL. THIS STATEMENT DOES NOT AFFIRM THAT THIS STUDENT WAS ENROLLED AS A STUDENT AT THE IE BUSINESS SCHOOL IN ANY WAY. IT DOES NOT CONFER A IE BUSINESS SCHOOL GRADE; IT DOES NOT CONFER IE BUSINESS SCHOOL CREDIT; IT DOES NOT CONFER A IE BUSINESS SCHOOL DEGREE; AND IT DOES NOT VERIFY THE IDENTITY OF THE STUDENT.



4 Courses

- Machine Learning Foundations: A Case Study Approach
- Machine Learning: Regression
- Machine Learning: Classification
- Machine Learning: Clustering & Retrieval

02/01/2017

Alexander Sommer

has successfully completed the online, non-credit Specialization

Machine Learning

Congratulations! This Certificate establishes that you have demonstrated proficiency in the exciting, high-demand field of Machine Learning through rigorous online coursework from leading Machine Learning researchers at the University of Washington. Through a series of practical case studies, you gained applied experience in major areas of Machine Learning including Prediction, Classification, Clustering, and Information Retrieval. You learned to analyze large and complex datasets, create systems that adapt and improve over time, and build intelligent applications that can make predictions from data. Take pride in your accomplishment and welcome to the global Machine Learning community!

Emily Fox, Amazon Professor of Machine Learning, Statistics

Carlos Guestrin, Amazon Professor of Machine Learning, Computer Science and Engineering

Verify this certificate at:
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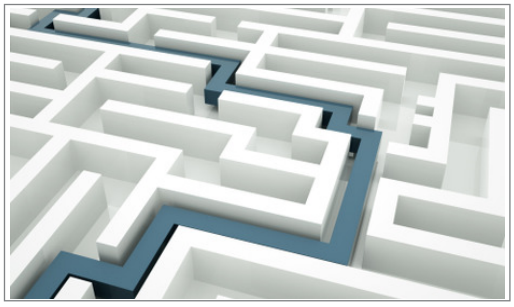
coursera.org

JUNE 08, 2015

Statement of Accomplishment

ALEXANDER SOMMER

HAS SUCCESSFULLY COMPLETED THE COURSERA ONLINE COURSE



Fundamentals of Project Planning and Management

This class introduces the concepts of project planning, Agile Project Management, critical path method, network analysis, and simulation for project risk analysis. Learners are equipped with the language and mindset for planning and managing successful projects.

Yael Grushka-Cockayne

YAEL GRUSHKA-COCKAYNE
ASSISTANT PROFESSOR OF BUSINESS ADMINISTRATION
DARDEN SCHOOL OF BUSINESS
UNIVERSITY OF VIRGINIA

IMPORTANT NOTE: THE ONLINE OFFERING OF THIS CLASS IS NOT IDENTICAL TO ANY COURSE OFFERED AT THE UNIVERSITY OF VIRGINIA ("UVA"). THE COURSERA PARTICIPANT WHO HAS RECEIVED THIS STATEMENT OF ACCOMPLISHMENT IS NOT ENROLLED AS A STUDENT AT UVA, HAS NOT RECEIVED CREDIT OR A GRADE FROM THE UNIVERSITY OF VIRGINIA, NOR HAS THE PARTICIPANT'S IDENTITY BEEN VERIFIED BY UVA.

COURSE CERTIFICATE

JANUARY 17, 2016

Alexander Sommer

has successfully completed

Machine Learning: Regression

a MOOC from the University of Washington and offered through Coursera



Emily Fox
Amazon Professor of Machine Learning
Statistics

Carlos Guestrin
Amazon Professor of Machine Learning
Computer Science and Engineering

Verify at coursera.org/verify/62JTKLY2JQSD

Coursera has confirmed the identity of this individual and their participation in the course.

deeplearning

COURSE CERTIFICATE

07/07/2019

Alexander Sommer

has successfully completed

Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning

an online non-credit course authorized by deeplearning.ai and offered through Coursera

Laurence Moroney

Laurence Moroney
Staff AI Advocate
Google Brain

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